

A Prompts

This appendix reproduces the prompt templates used in the pipeline verbatim. Placeholders in {braces} are filled in at runtime.

A.0.1 NER label generation (GPT-3.5-turbo). Used in the weakly-supervised labeling stage (Section 3.2) to convert paragraphs to concept lists for BIO conversion.

System.

You are a researcher with extensive knowledge about all the scientific fields.

User.

Output a python list of specific scientific concepts in the following text as valid lower case string encased within double quotes. Include acronyms and abbreviations as well.
{text}

Assistant prefix.

concept_list=

Sampling: temperature=0, top_p=1, max_tokens=512. The assistant prefix biases the completion to begin as a Python list literal, which is then parsed with `ast.literal_eval`.

A.0.2 Description system prompt (GPT-4o-mini). Shared by both description-generation modes (Section 3.5).

You are an expert scientific communicator. You have been tasked with summarizing information about topics, which you do very well. You try to summarize the topics using information from academic papers, but the people you are summarizing it for haven't read those papers – they only care about the topic.

A.0.3 Description prompt: topic-in-paper. Used when generating a topic description anchored to a specific paper (paper-detail page). The snippets are the top-*k* ranked ForeCite snippets for the (topic, paper) pair.

Here are text snippets selected from an academic paper, which are related to {topic_name}:
{snippets}

Write a summary about {topic_name} using solely these text snippets that would:

- Help someone understand what the {topic_name} is with respect to the snippets if they don't know about it already
- Be grounded in provided facts
- Focus on key relevant facts over author opinions or claims
- Exclude irrelevant facts that aren't about {topic_name}
- Take extra care to discern if something is a claim or a fact. Write: "The paper claims that {topic_name} is X Y Z" instead of "{topic_name} is X Y Z"
- Exclude facts that refer to external sources such as figures, references or other sources not included
- Exclude math facts
- Exclude citations

- Exclude numerical results
- The summary should be 1-3 sentences.
- If {topic_name} contains an acronym, make sure to expand it in the summary.
- The topic definition should be general enough to be true outside of the provided snippets. The snippets are from a specific paper, but the summary should be about the topic in {additional_context}, not just about the paper.

Output the summary as a json like this (include nothing else):

```
{
  "general_definition": "a general definition
of the topic (1 sentence)",
  "clarifying_information": "extra info that clarifies
the general_definition for someone unfamiliar
with the field and is more general than the
paper (1 sentence, optional)",
  "topic_in_paper": "what the topic is doing in
this paper (1 sentence, optional)"
}
```

The {additional_context} slot is general by default, switching to context of Computer Science when the topic does not occur in the paper's title or abstract *and* the topic has only a single defining paper (a heuristic that flags ambiguous, paper-specific surface forms).

A.0.4 Description prompt: topic-only. Used on the standalone topic page; conditioned on titles and abstracts of all defining papers rather than per-paper snippets. On context-length overflow the snippet list is iteratively halved (retry *i* keeps the first $|L| - 2^i$ entries) until the request succeeds.

Here are titles and abstracts selected from some academic papers, which are related to {title}:

{snippets}

Write a definition of {title}. The topic may have multiple definitions, but construct one that's related to the provided titles/abstracts. Some rules to keep in mind:

- Help someone understand what the {title} is if they don't know about it already
- There is no requirement to use the provided titles/abstracts in the definition – they are just there for context
- Exclude math facts
- Exclude citations
- Exclude numerical results
- If {title} contains an acronym, make sure to expand it in the summary.
- The definition should be 1-3 sentences in json format with the key as "definition".